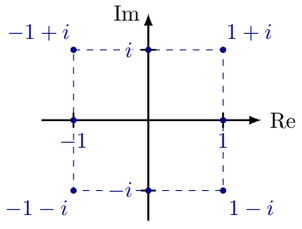
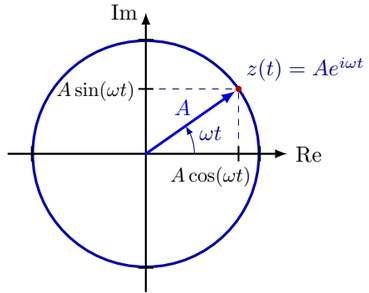
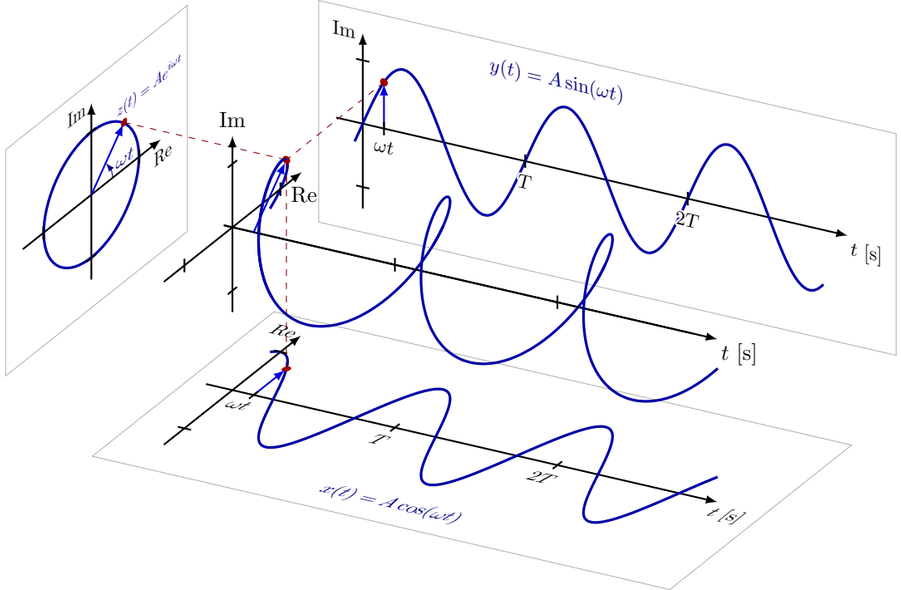
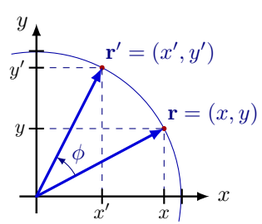
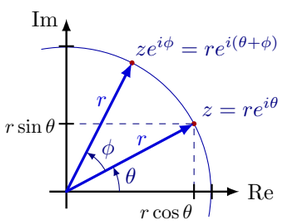

Im
Re
$z = x + iy = re^{i\theta}$
$\overline{z} = x - iy = re^{-i\theta}$
y
$-y$
x
r
θ
$-\theta$


Im
Re
$-1+i$
i
$1+i$
-1
1
$-1-i$
$-i$
$1-i$


Im
Re
$z(t) = Ae^{i\omega t}$
$A\sin(\omega t)$
$A\cos(\omega t)$
A
ωt


Im
Re
$z(t) = Ae^{i\omega t}$
ωt
$y(t) = A\sin(\omega t)$
T
$2T$
t [s]
$x(t) = A\cos(\omega t)$


y
y'
$\mathbf{r}' = (x', y')$
$\mathbf{r} = (x, y)$
ϕ
x'
x


Im
Re
$ze^{i\phi} = re^{i(\theta+\phi)}$
$z = re^{i\theta}$
r
ϕ
θ
$r\sin\theta$
$r\cos\theta$